

FRIDAY AUGUST 24th 2018 LECTURE 1

MAE 4730/5730, Intermediate Dynamics, Cornell Mechanical Engineering
Professor Andy Ruina.

VECTORS

- We will always have a fixed newtonian reference frame
- Newton's laws accurate $\approx 10^{-9}$

See Ruina/Pratap book chapter 1 about accuracy of Classical Mechanics.

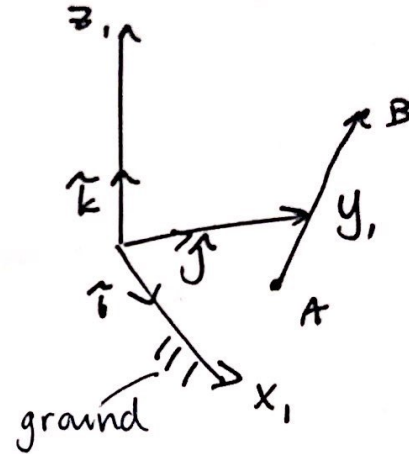


Reference frame:

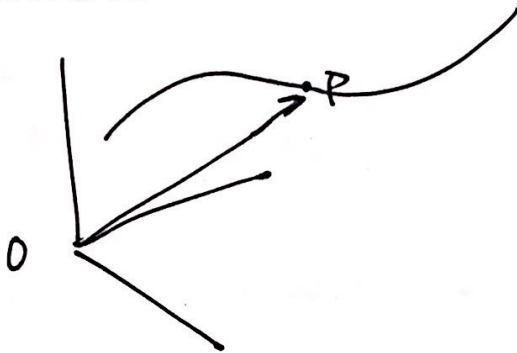
- Simply a coordinate system

$\hat{i}$ = unit vector

$\vec{r}_{AB} = \vec{r}_{B/A}$: position of B relative to A



Special case:



$$\vec{r}_{P/O} = \vec{r}_{O/P} = \vec{r}_P = r$$

shorthand

$$\vec{r} = r_x \hat{i} + r_y \hat{j} + r_z \hat{k} = x \hat{i} + y \hat{j} + z \hat{k} = \sum_{i=1}^3 x_i \hat{e}_i = x_i \hat{e}_i$$

$\sum_{i=1}^3 x_i \hat{e}_i = x_i \hat{e}_i$ in multiplicative expression if subscript appears twice $\Rightarrow \sum_i$

Velocity:

$$\vec{v}(t) = \frac{d\vec{x}}{dt} = \dot{\vec{x}} = \frac{d\vec{x}}{dt} = \dot{\vec{x}} = \vec{v} = \frac{d\vec{r}}{dt}$$

implied reference frame

$$\dot{\vec{r}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} \Rightarrow \Delta r = v \Delta t$$

$$\dot{\vec{x}} = \underbrace{x_i \dot{\hat{e}}_i + \dot{x}_i \hat{e}_i}_{\text{product rule}} = \dot{x}_i \hat{e}_i \equiv \frac{dF\vec{x}}{dt}$$

$\frac{d}{dt} \hat{e}_i = 0$ in newtonian

Acceleration:

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t}$$

$$\Delta v \approx \vec{a} \Delta t$$